Aristotle University of Thessaloniki
Electrical and Computer Engineering

Neural Networks Deep Learning

Semester Assignments 2026

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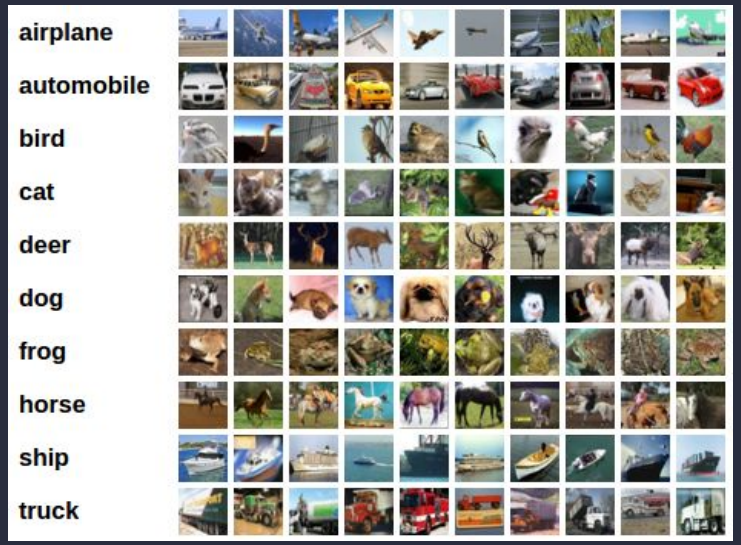
Contents

Datasets in use

CIFAR - 10

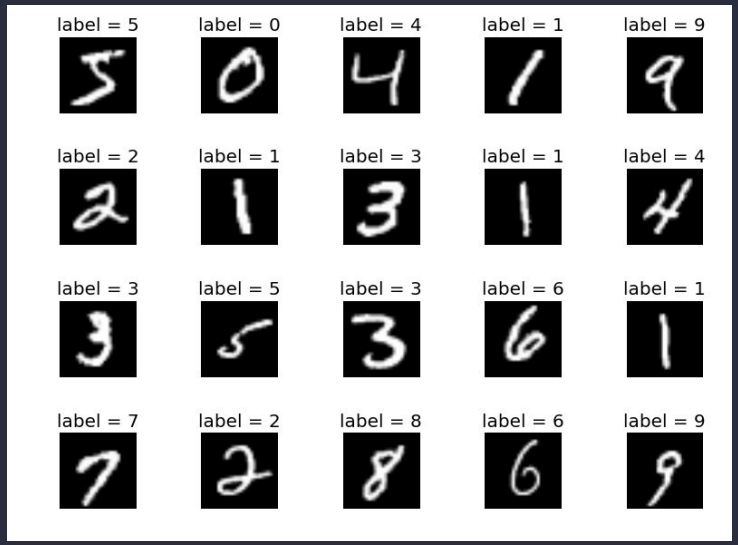


Dataset Size: 50,000 Training Set + **10,000** Test Set

Perfectly Balanced: 6,000 images per class

I**mage Dimensions: 32x32** pixels + 3 color channels (RGB)

MNIST

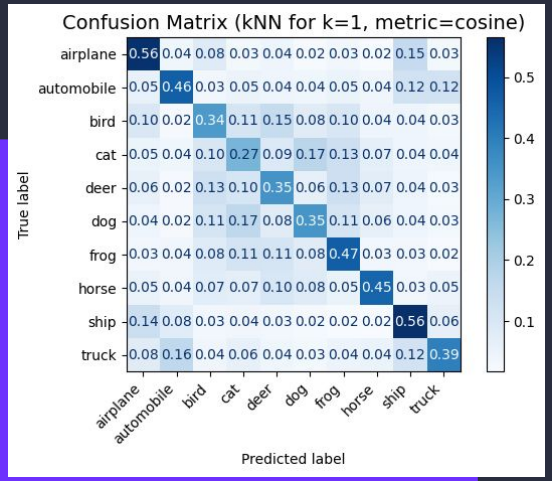


Dataset Size: 60,000 Training Set + **10,000** Test Set

Relatively Balanced: ~7,000 images per class

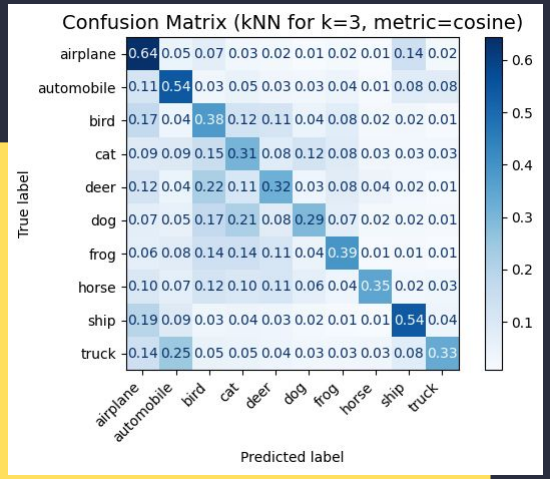
I**mage Dimensions: 28x28** pixels + 1 color channel (grayscale)

K - Nearest Neighbors



kNN for k=1, Cosine distance

Test accuracy: 42.11%



kNN for k=3, Cosine distance

Test accuracy: 40.97%

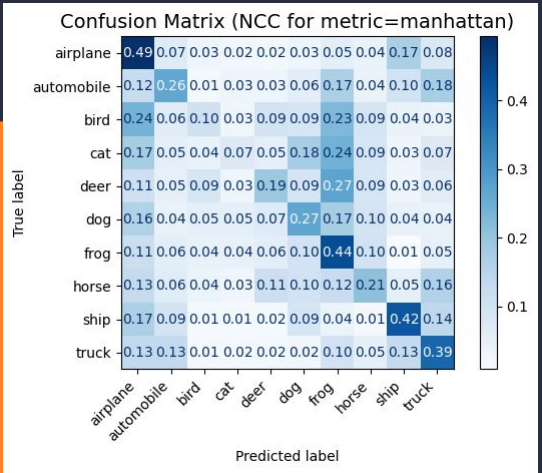
Distance Metrics tested:

- Euclidean
- Cosine
- Manhattan

Best case models for **Cosine** distance

PCA reduced dimensions to **50** for both kNN, NCC - Explained variance = **84.3%**

Nearest Class Centroid



Distance Metrics tested:

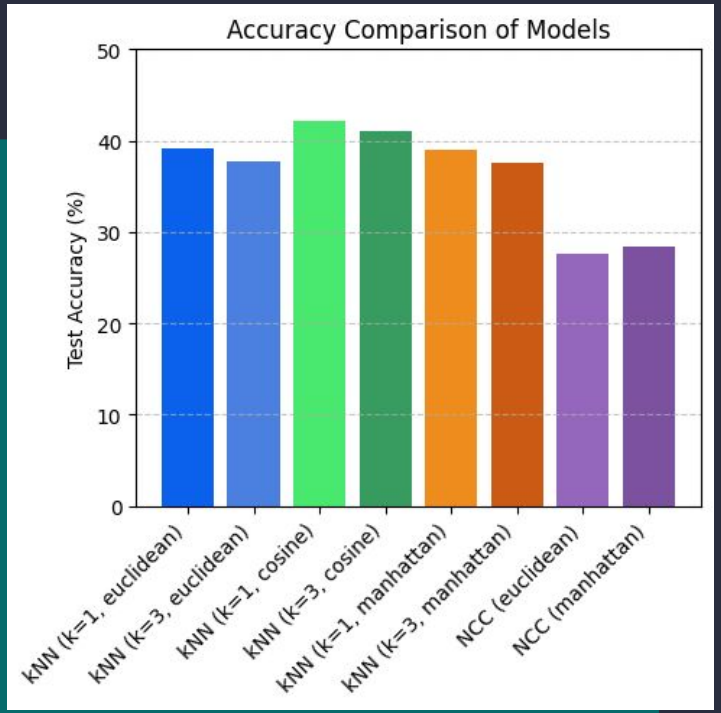
- Euclidean
- Manhattan

Best case model for **Manhattan** distance

NCC, Manhattan distance

Test accuracy: 28.47%

–> Classification reports were also printed for both kNN, NCC



Accuracy Comparison kNN + NCC

CNN

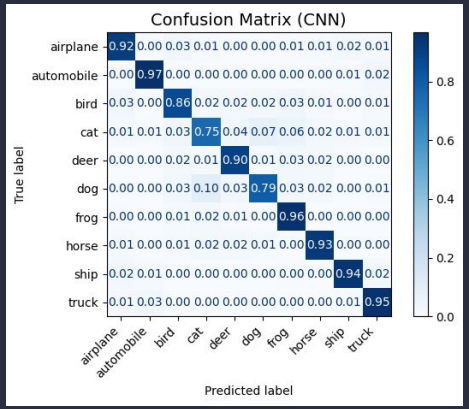
Best case model:
Accuracy **89.72%** - **57** epochs - Training Time **6521.26** s

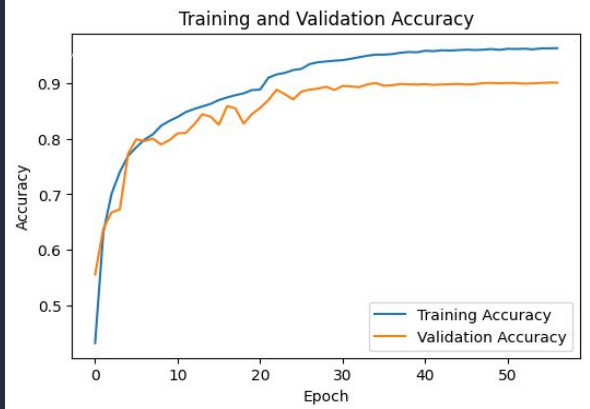
Layers: Conv2D (32, 64, 128, 256 filters), MaxPooling2D, BatchNormalization, GlobalAveragePooling2D, Dropout (0.3 only for output), Dense (for output layers)
Activation functions: Mostly "ReLu", "Softmax" for output layer
Optimizer: Adam
Loss: Sparse Categorical Crossentropy with Label Smoothing (custom)
Additional techniques: Data Augmentation, Early Stopping, Learning Rate Scheduler

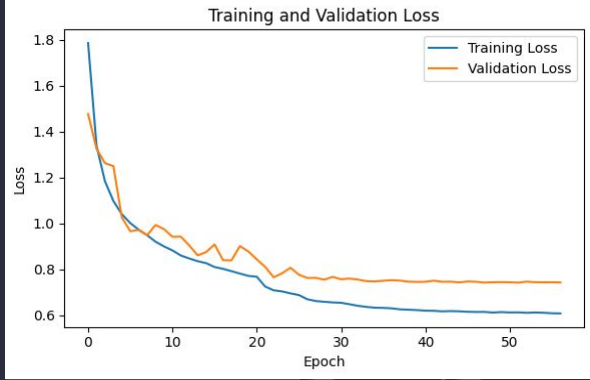
TensorFlow/Keras was used to build the model, which was optimized by:

- ➔ **Hyperparameter Tuning**
- ➔ **Trial-Error method**

Validation split was set to **0.2** for the training set, in order to keep test set untouched for the final evaluation of each model.







MLP

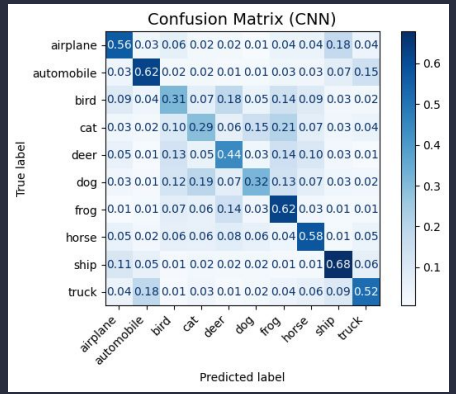
Random base model:
Accuracy **49.49%** - **50** epochs - Training Time **640.59** s

Layers: Dense (512, 256, 128, 10 for output), Flatten, BatchNormalization, Dropout (0.4)
Activation functions: Mostly "ReLu", "Softmax" for output layer
Optimizer: Adam
Loss: Sparse Categorical Crossentropy
Additional techniques: Data Augmentation, Early Stopping, Learning Rate Scheduler

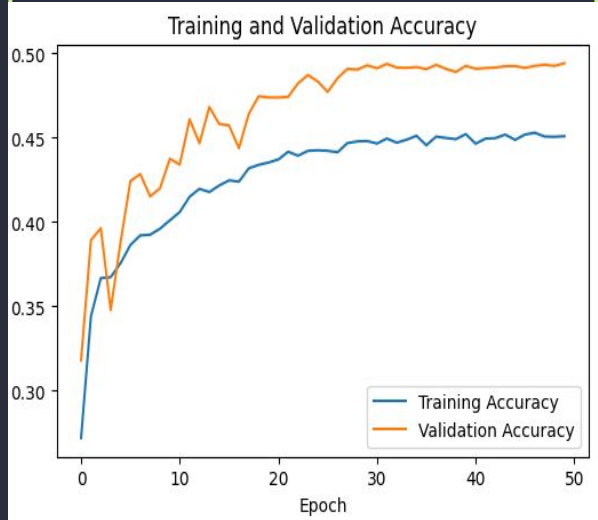
TensorFlow/Keras was used to build the model, based on:

- ➔ Best CNN hyperparameters and techniques

Validation split was set to **0.2** for the training set, in order to keep test set untouched for the final evaluation of each model.



We expect better performance on image datasets from CNNs, so a base MLP model was built only for comparison reasons:



SVM

2 Classes - Base

CIFAR-10 merged and balanced into 2 classes: Animals + Vehicles

Kernels: Linear, Polynomial, RBF, Sigmoid

Best model: RBF Kernel

- → **Hyperparameters:** C = 100, γ = 0.001
- → **Accuracy** = 88.37%

10 classes - Base

CIFAR-10 maintained with it's original classes

Kernels: Linear, Polynomial, RBF, Sigmoid

Best model: RBF Kernel

- → **Hyperparameters:** C = 1, γ = 0.01
- → **Accuracy** = 52.73%

2 Classes - Optimized

CIFAR-10 best 2-class SVM models optimized by additional techniques

Best model: RBF Kernel

- → **Hyperparameters:** C = 1, γ = 'scale'
- → Data Augmentation
- → Histogram of Oriented Gradients (HOG)
- → Color Histogram Features
- → **Accuracy** = 92.59%
- → Support Vectors Ratio = 31.33%

kNN - NCC - MLP

Used best SVM model techniques for comparison

Accuracy:

- → **kNN** (k=1): 84.74%
- → **kNN** (k=3): 87.38%
- → **NCC**: 81.76%
- → **MLP** (1 hidden layer, Hinge loss): 90.2%

Validation split was set to **0.4** for the training set, in order to keep test set untouched for the final evaluation.

Autoencoder

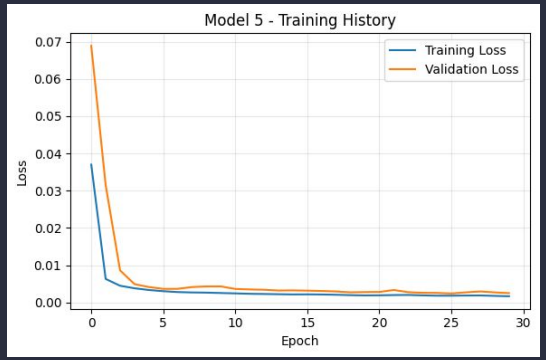
1. Digit Reconstruction

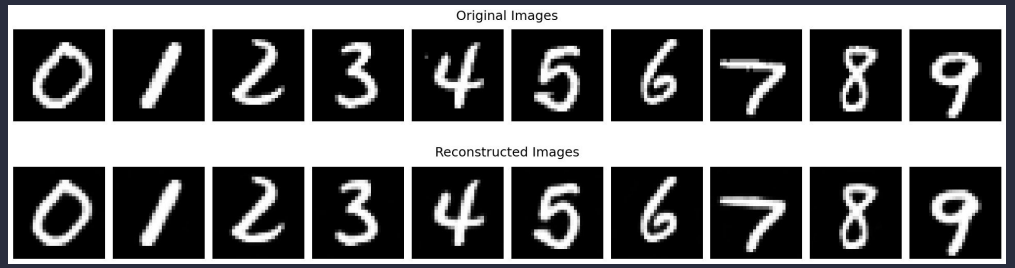
- **Model 1:** Dense layers (encoding dimension = 32)
- **Model 2:** Dense layers (encoding dimension = 64)
- **Model 3:** Dense layers (encoding dimension = 128)
- **Model 4:** Dense layers (encoding dimension = 128) + **PSNR, SSIM**
- **Model 5:** Convolutional model (encoding dimension = 512)

Best model: Convolutional

➔ **Test MSE:** 0.002452

➔ **Time:** 2705.37 seconds





Validation split was set to **0.4** (for all autoencoders) for the training set, in order to keep test set untouched for the final evaluation of each model.

Autoencoder

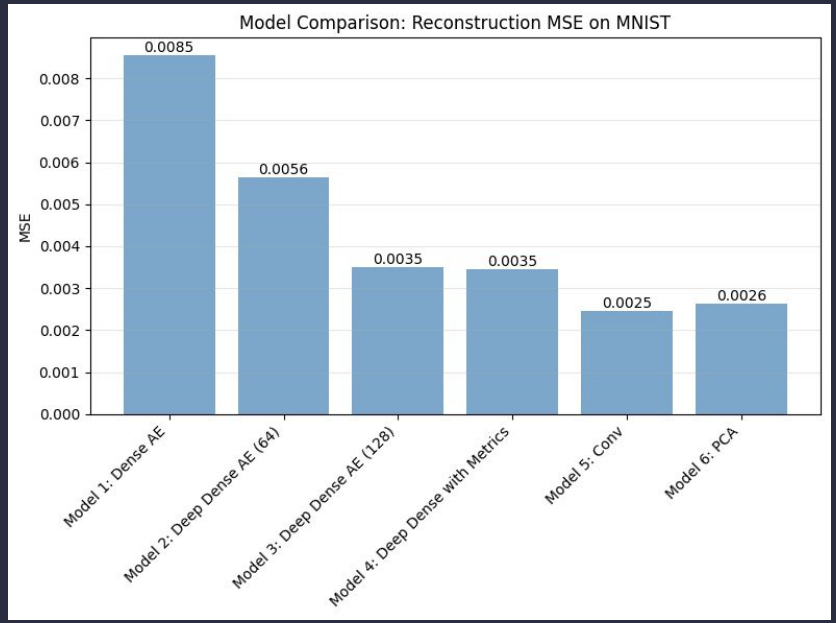
1. Digit Reconstruction

PCA Reconstruction for comparison: 784 -> 154 -> 784 (95% explained variance)

- ➔ PCA Test MSE: 0.002632
- ➔ Time: 4.86 seconds

CNN Classifier for reconstructed images:

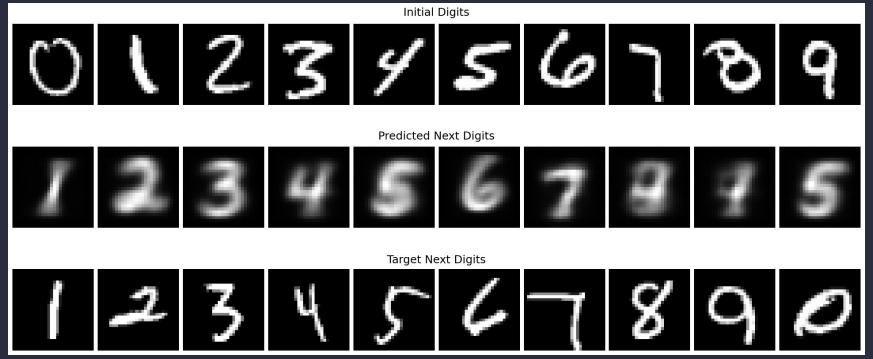
- ➔ Accuracy on Original Images: 0.9910
- ➔ Accuracy on Reconstructed Images: 0.9895

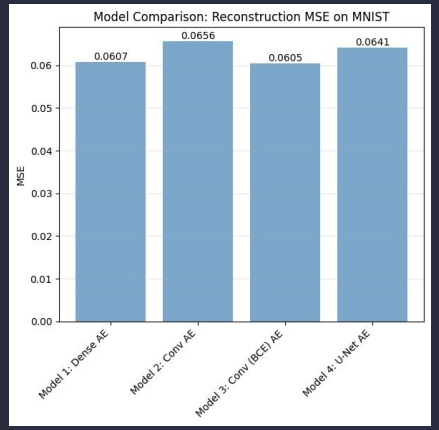


Autoencoder

2. Next Digit Reconstruction

- **Model 1:** Dense layers (encoding dimension = 128)
- **Model 2:** Convolutional model (encoding dimension = 512)
- **Model 3:** Convolutional model (encoding dimension = 512) + Binary Crossentropy Loss **(BCE)**
- **Model 4:** U-Net model with **Skip Connections** + Learning Rate Scheduler



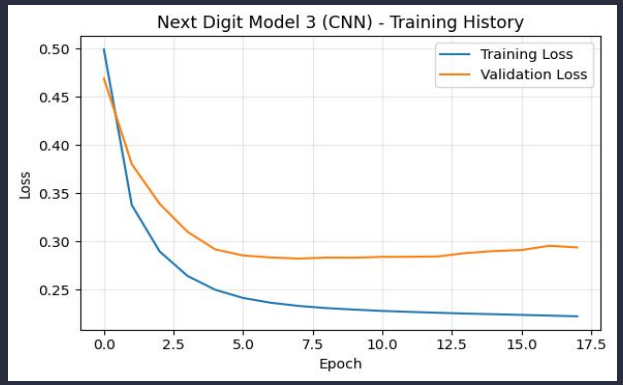


Best model: Convolutional + BCE

➔ **Test MSE:** 0.060738

➔ **Time:** 1925.83 seconds

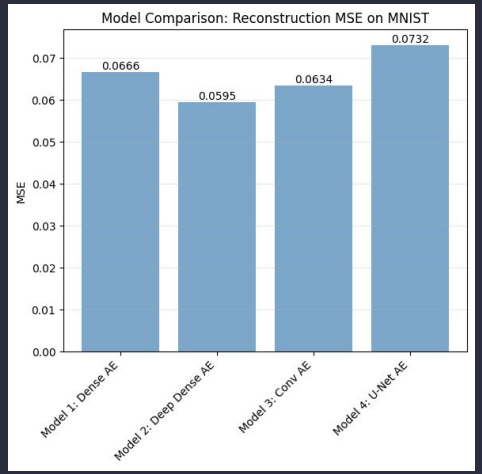
Model still needs improvement, last digits are wrongly reconstructed



Autoencoder

3. Digit Addition Reconstruction

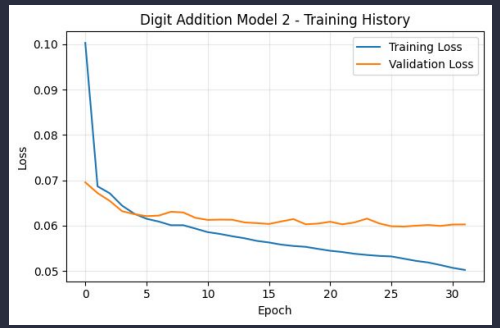
- **Model 1:** Dense layers (encoding dimension = 256)
- **Model 2:** Dense layers (encoding dimension = 512)
- **Model 3:** Convolutional model
- **Model 4:** U-Net model with **Skip Connections** + Learning Rate Scheduler

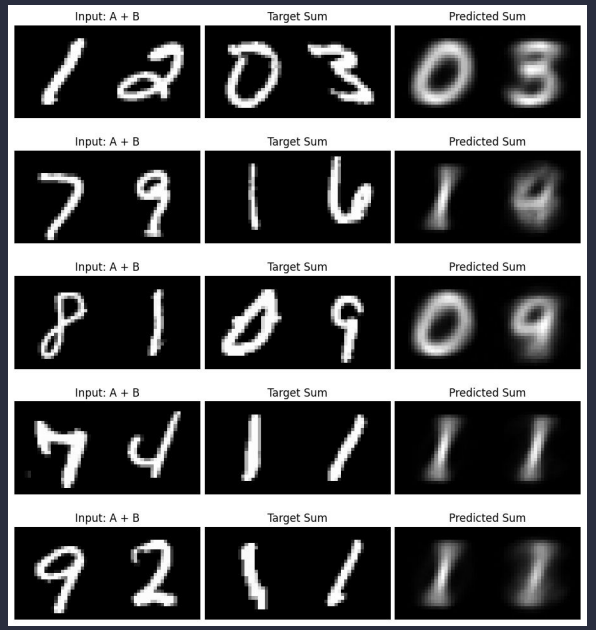


Best model: Dense (512)

➔ **Test MSE:** 0.059476

➔ **Time:** 177.88 seconds





Skip Connections failed due to:
Encoder input ≠ Decoder output

Comparison results

- **CNN vs. MLP:** CNN tend to achieve much higher accuracy compared to MLP because **spatial layers** (Conv2D) are superior to MLP's **flattened** approach for image data, though training times are significantly different.
- **SVM:** The SVM performance dropped significantly for **2 classes** compared to **10 classes** , proving that SVMs work best for a few classes where clear margins can be established.
- **Digit Reconstruction:** While **Convolutional Autoencoder** provided very accurate reconstruction, **PCA** reconstruction was nearly as effective and vastly more efficient.
- **Next Digit & Addition:** These complex tasks yielded higher error rates and proved that not all advanced architectures e.x. **Skip Connections** are suitable for tasks like these (Encoder input ≠ Decoder output).

Thank you

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